

FSP Commissioning and Modal Fitting in a New Environment

A repeatable workflow for plant calibration, open-loop modal identification, normal-loop qualification, and push-event validation

Anyloop levfork / parallel DAC configuration • Prepared 2026-08-01 • Applies to steering_par_fsp.json, attenuation_par_fsp.json, and push_event_par_fsp.json

Core principle. Do not ask one experiment to identify everything. The actuator Bode tests identify the command-to-centroid plant. The attenuation run's open phase identifies the environmental vibration model. The push run validates transient rejection after both models are installed.

1	2	3	4	5	6
Full x/y Bodes	Normal steering smoke test	Full attenuation run	Fit open-loop modes	Update all configs	Push-event validation

1. Parameter provenance

Parameter	Meaning	Correct source
K	Signed command-to-error gain used for voltage scaling and command-echo subtraction.	Actuator Bode; prefer the lower-amplitude pass for operating-point gain.
delay, delay_frac	Total command-to-measurement delay in frames.	Full-band x/y Bodes, never a narrow small-signal sweep.
plant_b/a	Optional residual command-path dynamics beyond gain and delay.	Complex full-band actuator Bode fit.
freqs	Environmental modal center frequencies.	Open-loop centroid PSD; verify with command-subtracted push ring-downs.
zeta	Modal damping ratio controlling pole radius and linewidth.	Resolved open-loop PSD or released ring-down decay.
q	AR(2) modal drive variance—not visible	Open-loop modal variance

	peak height or mechanical stiffness.	divided by FSP's stationary variance gain G_v .
r	Measurement/unmodeled-disturbance variance reference.	Residual open-loop spectrum. The fitter normalizes it to 1.
transient_kp	Dimensionless fraction of instantaneous error corrected by the event P path.	Controller validation. Effective command gain is already K_p/K .
$\text{transient_modal_q_scale}$	Event observer aggressiveness relative to the stationary modal model.	Push validation; begin at 100.

Kp scaling: the event term is $u_P = -(K_p \times \text{error})/K$. Updating measured K automatically updates

volts per unit error. Multiplying configured K_p by K would defeat this normalization.

2. Stage 1 — actuator calibration

1. Center the beam and retain the ROI, exposure, gain, bandwidth, DAC mapping, bias, and supplies intended for steering.
2. Run `conf_bode_par_x.json` and `conf_bode_par_y.json`.
3. Update each axis's K and delay. Update `plant_b/a` together if a residual biquad is fitted.
4. Use the lower-amplitude pass's low-frequency plateau for initial K when the two amplitudes disagree. Use the full sweep for delay.

Large Bodes first is acceptable for commissioning, but not final optimization. This FSM has shown

several-percent compression. A configured K below the true small-signal gain over-commands. Until

small sweeps confirm K, keep pushes modest and watch the burst guard.

Stop and remeasure for a wrong sign, implausible delay, strong amplitude disagreement, ROI

clipping, or measured sample rate inconsistent with configured f_s .

3. Stage 2 — ordinary steering smoke test

Run `steering_par_fsp.json` without intentional pushes. This qualifies the continuously adaptive broadband predictor, command echo, clamps, burst guard, beam-loss behavior, and autoranging under ordinary disturbance.

Go/no-go gate

- ☐ Beam stays in the tracked ROI; no beam-loss hold.
- ☐ Commands remain inside the normal ± 1 V envelope.
- ☐ No recurring burst-guard activation or narrow high-frequency command line.
- ☐ No unexplained transient after the arm delay.
- ☐ Error and command remain finite and visually stationary.

With `broad_order > 0`, normal prediction uses the broadband FIR rather than the event modal model. The plant can therefore be qualified before new modal arrays exist.

4. Stage 3 — attenuation and open-loop capture

Run the complete `attenuation_par_fsp.json`. Its pre-FSP sink writes `atten_par_err6.aylp`. During the attenuation test's open interval, the DAC is parked and the recorded centroid is the environmental disturbance directly. `pass_open=false` prevents open data from training FSP but does not change the saved measurement.

Go/no-go gate

- ☐ Open phase has no intentional push, ROI clipping, or prolonged beam loss.
- ☐ Wiener trace converges before the scored closed window.
- ☐ Closed/open RMS and bands show useful rejection.
- ☐ No recurring guard, command warning, or unexplained transient.
- ☐ Shadow promotions are zero in a clean stationary run.

Do not fit a damaged open interval. Median-Welch processing rejects an isolated touch, but not a

beam pinned at the ROI edge or a long environmental change. Repeat the run or explicitly select a

clean subsection.

5. Stage 4 — fit freqs, zeta, and q/r

```
python3 contrib/fit_fsp_modes.py atten_par_err6.aylp \  
--config contrib/attenuation_par_fsp.json \  
--output fitted_fsp_modes.json
```

The script reads `fs`, `start_delay`, and `open_time` automatically. It reads only that interval, computes a long-segment median-Welch PSD with GSL, detects prominent lines, fits a Lorentzian around each, converts linewidth to zeta, converts modal power to FSP's AR(2) drive `q`, normalizes residual variance to `r=1`, and respects the eight-mode axis limit.

Fit model: $\text{PSD}(f) = B + A/[1 + ((f-f_0)/\text{gamma})^2]$, with $\text{zeta} \approx \text{gamma}/f_0$ and modal variance $\approx \pi A \text{ gamma}$.

Useful options

Option	Purpose
<code>--start-s</code> <code>--duration-s</code>	Override automatic selection for a known clean subsection.
<code>--segment-s 90</code>	Damping resolution versus number of independent averages.
<code>--fmin 4 --fmax 150</code>	Candidate modal band.
<code>--prominence-db 4</code>	Minimum elevation over the local spectral floor.
<code>--min-q-r 1e-7</code>	Reject energetically irrelevant noise peaks.
<code>--max-modes 8</code>	Controller state budget; never exceed 8 per axis.

Review before copying

- Prefer modes that are prominent and carry meaningful fitted power.
- `zeta resolution-limited` is a bound, not a precise measurement. Retain a conservative prior or use released ring-down decay.
- Do not keep duplicate fits of one line or round close modes into one.
- Tiny q/r modes add state without useful control authority.
- The fitter deliberately does not rewrite production JSON.

6. Stage 5 — install one consistent model

Copy the reviewed `r`, `freqs`, `zeta`, and `q` fragments into all three files:

- `contrib/attenuation_par_fsp.json`
- `contrib/steering_par_fsp.json`
- `contrib/push_event_par_fsp.json`

Array lengths must match. Retain `transient_modal_q_scale=100` and `transient_kp=0.15` for the first comparison; these are recovery settings, not identification outputs.

Configuration invariant: K, delay, optional plant biquad, fs, and modal arrays must describe the

same alignment and transport. Re-centering, changing camera settings, changing DAC mapping, or

moving the apparatus can invalidate the model.

7. Stage 6 — released-push validation

Use `push_event_par_fsp.json`, not ordinary steering, for intentional pushes. It retains synchronized pre-FSP error, actual post-clamp command, and a CSV row per completed event.

1. Wait through startup, authority ramp, and detector warm-up.
2. Apply at least six cleanly released pushes mainly along y and six mainly along x.
3. Begin around 3–6 pixels—not near the ROI boundary merely because event authority can reach ± 10 V.
4. Leave at least 10 seconds after recovery before the next push.
5. Avoid diagonal pushes where possible; event number and A/B parity are per axis.

The supplied push config has `transient_modal_ab=true`: odd events use proportional recovery and even events modal recovery. This is the best same-run evidence that modal recovery earns its keep. Set false only for modal-only qualification.

Analyze recovery and mode validity

```
python3 contrib/analyze_push_events.py push_modal_events.csv \
--frequency-test \
--error-aylp push_modal_error.aylp \
--command-aylp push_modal_command.aylp \
--config contrib/push_event_par_fsp.json
```

$$\text{disturbance} = \text{measured error} - K H(z) D(z) \text{ applied command}$$

Subtracting the actual delayed command makes different pushes comparable. Push strength changes modal amplitude; frequency median and event-to-event MAD are the invariants.

Analyzer result	Meaning	Action
PASS	Enough pushes excited the mode; median and MAD meet tolerance.	Keep it.
UNTESTED	Too few pushes visibly excited it.	Collect more; do not delete it.
Shifted / FAIL	A nearby strong line repeatedly falls outside tolerance.	Review frequency, update, rerun.
Repeatable residual	A strong line is absent from accepted configured modes.	Add it or replace the weakest mode.
Low coverage	Ring-down is broadband, nonlinear, or incompletely modeled.	Inspect residuals and beam validity; do not blindly raise q-scale.

Shadow promotion	A persistent regime passed private shadow-NLMS validation.	Investigate environmental change; ordinary pushes should not promote.
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8. Final acceptance criteria

Normal loop

- ☐ Better closed/open attenuation in important bands.
- ☐ Broadband weights converge before scoring.
- ☐ Normal commands stay within ± 1 V.
- ☐ No recurring burst guard or parasitic line.
- ☐ Only beam loss holds output at zero; reacquisition resumes operation.

Transient loop

- ☐ Events appear in the terminal and CSV.
- ☐ Modal recovery improves median settling and/or RMS without unacceptable peak command.
- ☐ Excited frequencies pass validation.
- ☐ No strong repeatable residual remains.
- ☐ No event saturation, ROI loss, guard conflict, or unexplained shadow promotion.

Decision rule. Keep modal recovery only if it improves push recovery on the physical rig while ordinary attenuation remains sound. Require repeatability across different push sizes and command-subtracted frequency stability—not one visually good event.

9. Terminal messages to act on

Message	Interpretation
<code>transient recovery on ...</code>	Event threshold crossed; selected recovery and Kp are printed.
<code>transient quiet; cross-fading ...</code>	Disturbance returned to the known-model envelope.
<code>event shadow model promoted ...</code>	A persistent new predictable regime passed validation.
<code>burst guard tripped ...</code>	Likely K/delay mismatch or regeneration; re-run Bodes.
<code>command/error limit ...</code>	Diagnostic exceeded; inspect requested/clamped command and beam position.
Beam lost / reacquired	Zero is held only during loss; dynamic state is cleared and authority ramps back.

Implementation references: `contrib/fit_fsp_modes.py`, `contrib/analyze_push_events.py`, `doc/devices/fsp.md`, and the parallel-DAC FSP configurations named above.